

MA22015 2026 - Coursework

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Generative AI statement

```
# Type B: Generative AI is permitted as an assistive tool within the assessment.
# GenAI was used in the following ways:
# - To write code to extract and neatly output relevant information from Linear
#   model summaries to keep output concise and avoid outputting full summaries
# - To find the pairs() function to visualise correlation in question 6
# - To proof read my answers
# - To find ifelse() function to tidy my code on question 5
```

```
# Further note:
# - I had various issues with R which I discussed in my lab. I was instructed:
#   - use xelatex instead of tinytex
#   - I had an issue with special characters in the data, calling
#     Sys.setlocale("LC_CTYPE", "en_US.UTF-8") fixed it.
# - Code has been used and referenced from data science module throughout
```

Setup

```
polpres <- read.csv("polpres.csv")

# used from Data science module
library(tidyverse)
library(knitr)
library(janitor)
```

The wordcount:

```
library(rmdwc)
```

Warning: package 'rmdwc' was built under R version 4.4.3

```
rmdcount("cw26.Rmd")$words
```

```
[1] 1491
```

1. Additional Voters

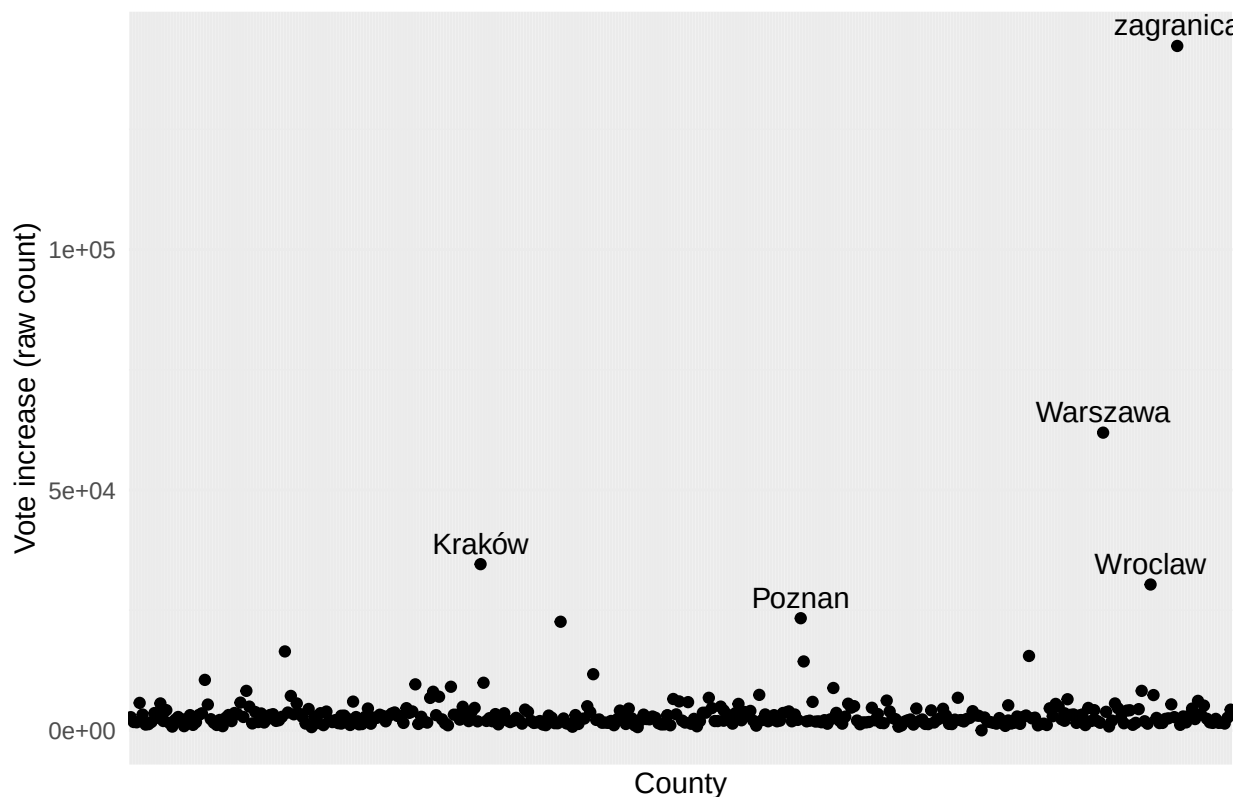
```
polpres <- polpres %>%
  mutate(
    r1_votes = Bartoszewicz + Biejat + Braun + Hołownia + Jakubiak +
      Maciak + Mentzen + Nawrocki + Senyszyn + Stanowski +
      Trzaskowski + Woch + Zandberg,
    r2_votes = spoilt + N2 + T2)

polpres <- polpres %>%
  mutate(r2_voter_increase = r2_votes - r1_votes)

top_5_vote_change <- polpres %>%
  select(county, r2_voter_increase, r1_votes, r2_votes) %>%
  slice_max(order_by = abs(r2_voter_increase), n = 5) %>%
  arrange(desc(r2_voter_increase))

polpres %>%
  ggplot(aes(x = county, y = r2_voter_increase))+
  geom_point()+
  geom_text(
    data = top_5_vote_change,
    aes(label = county),
    vjust = -0.5)+
  theme_minimal()+
  theme(axis.text.x = element_blank())+
  labs(x = "County", y = "Vote increase (raw count)",
    title = "Increase/decrease in voters round 1 to round 2")
```

Increase/decrease in voters round 1 to round 2



I created “r2_voter_increase” with the increase of voters for each county in r2. There seemed to be unusual changes: “Zagranica” increased by 142347 voters. “Zagranica” and “Warszawa” both have large populations, so I computed % increase as raw counts are biased to bigger populations.

```
polpres <- polpres %>%
  mutate(pct_r2_voter_increase = 100 * (r2_votes - r1_votes) / r1_votes)

top_5_pct_vote_change <- polpres %>%
  select(county, pct_r2_voter_increase, r1_votes, r2_votes) %>%
  slice_max(order_by = abs(pct_r2_voter_increase), n = 5) %>%
  arrange(desc(pct_r2_voter_increase))

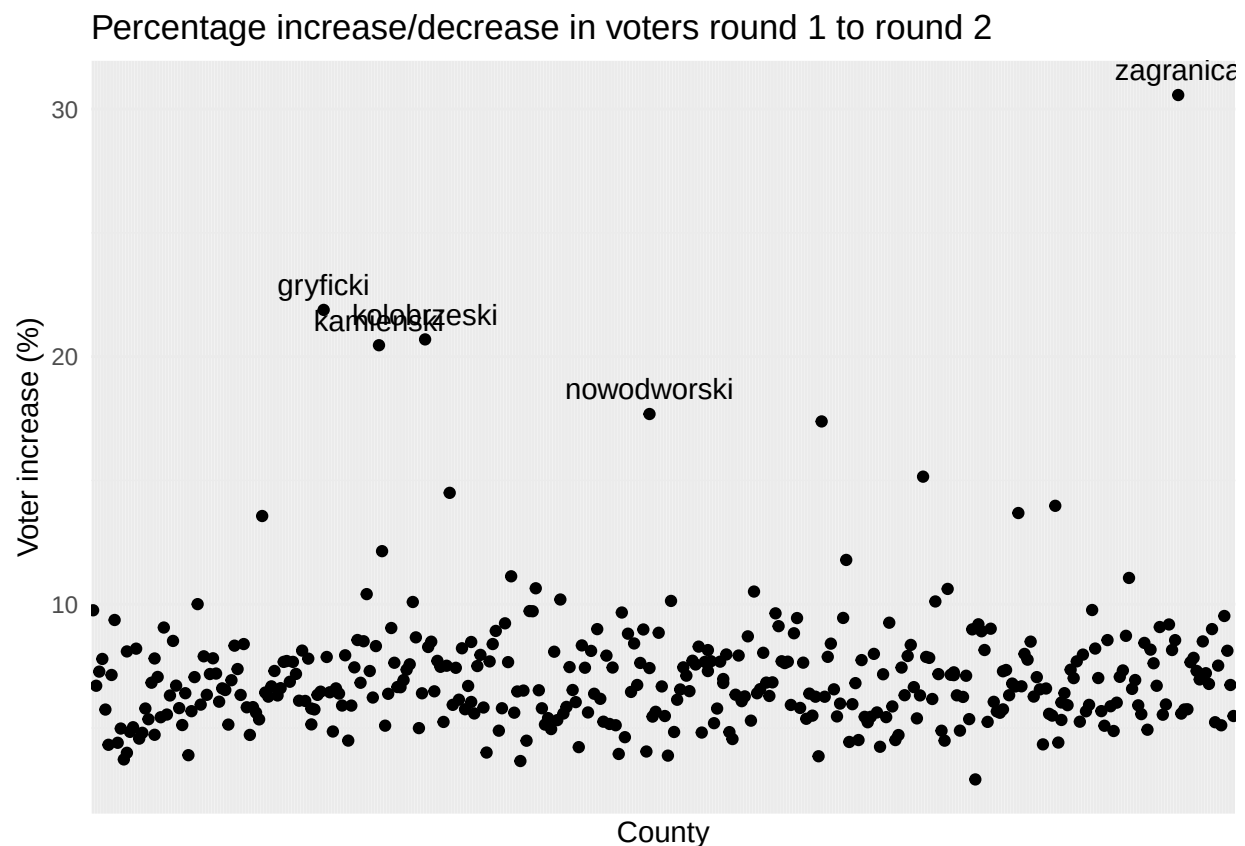
top_5_pct_vote_change %>%
  kable(caption = "Top 5 counties by percentage change in No. of voters")
```

Table 1: Top 5 counties by percentage change in No. of voters

county	pct_r2_voter_increase	r1_votes	r2_votes
zagranica	30.56651	465696	608043
gryficki	21.88804	27330	33312
kołobrzeski	20.69634	43829	52900
kamieński	20.45998	22566	27183
nowodworski	17.68255	17571	20678

```
# kable() from knitr has been used throughout, for neat table outputs in our compiled file
# Referenced from Data science module
```

```
polpres %>%
  ggplot(aes(x = county, y = pct_r2_voter_increase))+
  geom_point()+
  geom_text(
    data = top_5_pct_vote_change, # geom_text() used from data science
    aes(label = county),
    vjust = -0.7)+
  theme_minimal()+
  theme(axis.text.x = element_blank())+
  labs(x = "County", y = "Voter increase (%)",
       title = "Percentage increase/decrease in voters round 1 to round 2")
```



The majority of counties have a 5-10% increase in r2 voters. Again “Zagranica” has a disproportionate increase of 31%. Gryficki, kolobrzescski and kamieniski also have large increases over 20%.

```
lmod_vote_increase <- lm(pct_r2_voter_increase ~ 1, data = polpres)
polpres_resid <- polpres %>%
  mutate(student_resid = rstudent(lmod_vote_increase))
polpres_resid %>%
  arrange(desc(abs(student_resid))) %>%
  select(county, student_resid) %>%
  head(7) %>%
  kable(caption = "Studentised residuals by County")
```

Table 2: Studentised residuals by County

county	student_resid
zagranica	10.288674
gryficki	5.992483
kołobrzesci	5.468267
kamieński	5.365617
nowodworski	4.187751
pucki	4.063733
ślawieński	3.151966

```
ncounties <- nrow(polpres)
vote_increase_cv <- abs(qt(.05/(ncounties*2),ncounties-1-1))

cat("Bonferroni critical value is", round(vote_increase_cv,3))
```

Bonferroni critical value is 3.864

I conducted a test to see which counties were outliers assuming all counties have the same mean increase. Six observations exceeded the Bonferroni critical value (3.864), so these are statistically outliers relative to the overall mean percentage increase. Zagranica had the largest value, indicating it was the largest outlier.

I looked closer at Zagranica:

```
polpres %>%
  filter(county == "zagranica") %>%
  select(ccode, county, eligible) %>%
  kable()
```

ccode	county	eligible
NA	zagranica	524204

Zagranica doesn't have a county code. It translates to "abroad", so we cannot directly compare its behaviour to other counties as it is a fundamentally different group.

To conclude, among "counties" there is still evidence of unusually large vote increases in gryficki, kołobrzesci, kamieński, nowodworski and pucki.

2. Exit Poll

```
proportion_T2 <- sum(polpres$T2) / (sum(polpres$T2) + sum(polpres$N2))
cat("Trzaskowski received", round(proportion_T2*100, 1), "% of the votes in round 2.", "\n")
```

Trzaskowski received 49.1 % of the votes in round 2.

```
# genAI used to find \n to put it on a new line
```

```
p <- 1 - pbinom(500, size = 1000, prob = proportion_T2)
cat("Probability of majority in random sample of 1000:", round(p,2), "\n")
```

Probability of majority in random sample of 1000: 0.28

The r2 proportion for Trzaskowski was 49.1%, so a randomly selected voter would have voted for him with estimated probability 0.491.

Assuming randomness, X votes for Trzaskowski in 1000 voters is $\sim \text{binom}(1000, 0.491)$, so probability that the majority voted Trzaskowski is $1 - p(X \leq 500) = 0.28$.

Around 1 in 4 exit polls would incorrectly show Trzaskowski leading, so an early exit poll showing Trzaskowski ahead would not be surprising even if Nawrocki ended as the overall winner.

3. Turnout

```
polpres %>%
  filter(is.na(ccode)) %>%
  select(ccode, county, province, eligible) %>%
  kable(caption = "Closer inspection of missing ccodes")
```

Table 4: Closer inspection of missing ccodes

ccode	county	province	eligible
NA	statki		173
NA	zagranica		524204

```
polpres_polish_citizens <- polpres %>%
  filter(county != "zagranica") %>%
  filter(county != "statki") %>%
  mutate(r2_pct_turnout = 100 * (spoilt + N2 + T2) / eligible2)
```

Filtering voters registered in Poland:

From q1, “zagranica” refers to abroad voters. I checked which other counties are missing ccodes: I found “statki”, translating to “ships”. I removed these.

Pct turnout for the remaining counties:

```
turnout_province <- polpres_polish_citizens %>%
  group_by(province) %>%
  summarise(Mean = mean(r2_pct_turnout),
            SD = sd(r2_pct_turnout)) %>%
  arrange(desc(Mean))

turnout_province %>%
  kable(caption = "Percentage turnout by Province (excluding voters not registered in Poland)")
```

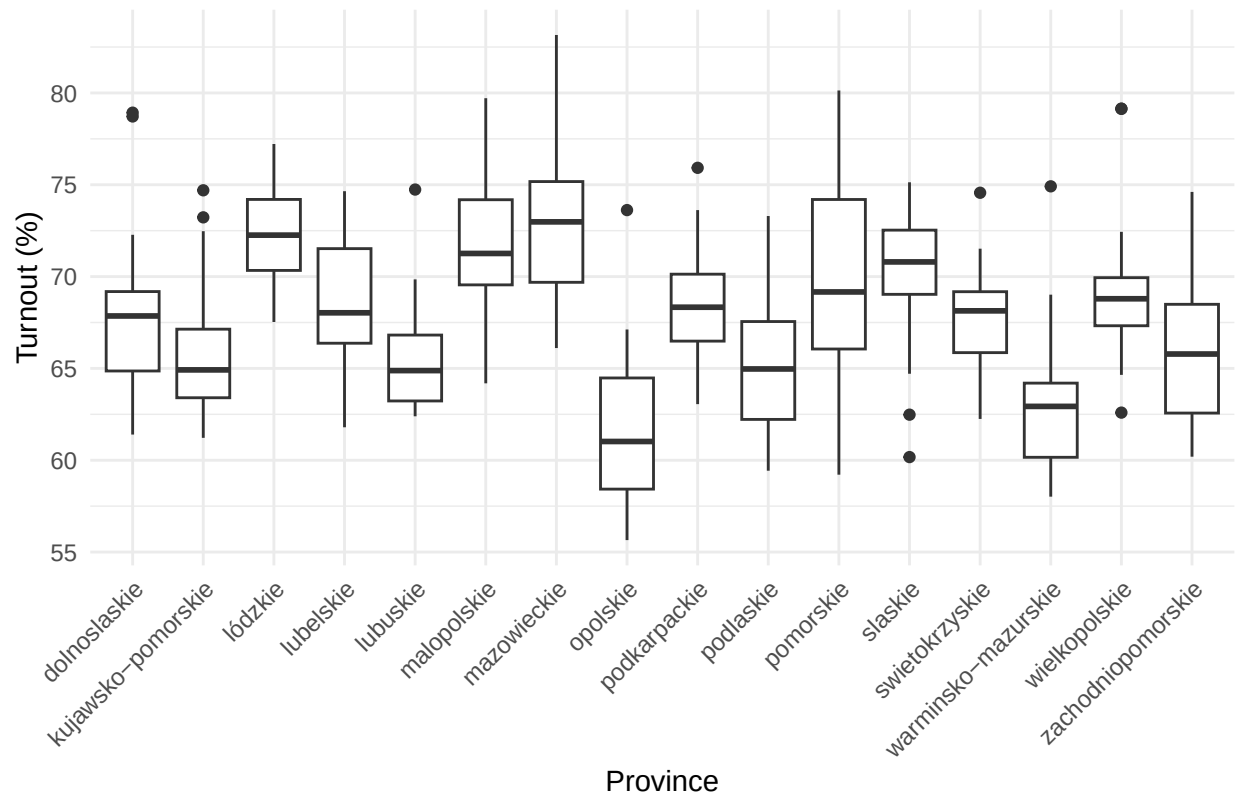
Table 5: Percentage turnout by Province (excluding voters not registered in Poland)

province	Mean	SD
mazowieckie	73.23025	4.715780
łódzkie	72.22566	2.539700
małopolskie	71.88063	3.637431
śląskie	70.24083	3.504920
pomorskie	69.90870	5.834419
wielkopolskie	69.06677	3.222075
lubelskie	68.58240	3.651375
podkarpackie	68.45189	2.982185
świętokrzyskie	67.89386	3.039137
dolnośląskie	67.69631	4.134894
zachodniopomorskie	66.68675	4.714002
kujawsko-pomorskie	65.81981	3.698078
lubuskie	65.72652	3.351291
podlaskie	65.32562	4.106631
warmińsko-mazurskie	63.03296	3.954050
opolskie	61.96354	5.094410

The means ranged from 61-74%. I made a plot to visualise:

```
polpres_polish_citizens %>%
  ggplot(aes(x = province, y = r2_pct_turnout))+
  geom_boxplot()+
  labs(x = "Province", y = "Turnout (%)",
       title = "Percentage turnout by Province (excluding voters not registered in Poland)")+
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) # used from data science
```

Percentage turnout by Province (excluding voters not registered in Poland)



module to neaten labels

There is moderate variation between provinces, but no single Province is striking as exceptionally high or low. Mazowieckie and opolskie have the highest and lowest mean turnouts respectively.

I decided to do an anova test to see if the difference in provinces is statistically significant.

```
lmod_null <- lm(r2_pct_turnout ~ 1, data = polpres_polish_citizens)
lmod_turnout <- lm(r2_pct_turnout ~ province, data = polpres_polish_citizens)
anova(lmod_null, lmod_turnout)$`Pr(>F)`[2]
```

```
[1] 8.686333e-29
```

genAI has been used to extract relevant parts of the summary, to keep the output concise

We get an extremely small p-value: strong evidence the turnout depends on the Province, and we do not have equal means.

I checked whether any provinces are outliers:

```
lmod_province <- lm(Mean ~ 1, data = turnout_province)
turnout_province <- turnout_province %>%
  mutate(student_resid = rstudent(lmod_province))
turnout_province %>%
  arrange(desc(abs(student_resid))) %>%
```

```
select(province, student_resid) %>%
head(5) %>%
kable(caption = "Studentised residuals by Province")
```

Table 6: Studentised residuals by Province

province	student_resid
opolskie	-2.198375
mazowieckie	1.841321
warmińsko-mazurskie	-1.714557
łódzkie	1.430029
małopolskie	1.299005

```
bonferroni_critical_value <- abs(qt(.05/(16*2),16-1-1))

cat("Bonferroni critical value is", round(bonferroni_critical_value,3))
```

Bonferroni critical value is 3.562

I fitted a linear model under the assumption that turnout doesn't depend on the province. I computed the studentized residuals, and the Bonferroni critical value (0.05 significance, 14df) was 3.562. Opolskie and mazowieckie had the most extreme residuals (consistent with our earlier plot) however 2.19 does not exceed 3.562 so it is not an outlier.

Although the anova test shows there is statistically significant variation in between provinces, no individual province stands out as exceptionally high or low turnout.

4. Spoilt Ballots

```
polpres_spoilt_votes <- polpres %>%
  mutate("T2_proportion" = T2/(N2 + T2)) %>%
  select(county, province, eligible2, spoilt, N2, T2, T2_proportion) %>%
  mutate("pct_spoilt" = 100* (spoilt/(N2+T2+spoilt)))

polpres_spoilt_votes %>%
  arrange(desc(pct_spoilt)) %>%
  slice_head(n=5) %>%
  select(county, pct_spoilt) %>%
  kable(
    caption = "Percentage of votes spoilt vs proportion of votes for \n Trzaskowski in each County (roun
```

Table 7: Percentage of votes spoilt vs proportion of votes for Trzaskowski in each County (round 2)

county	pct_spoilt
statki	2.272727
sztumski	1.157315

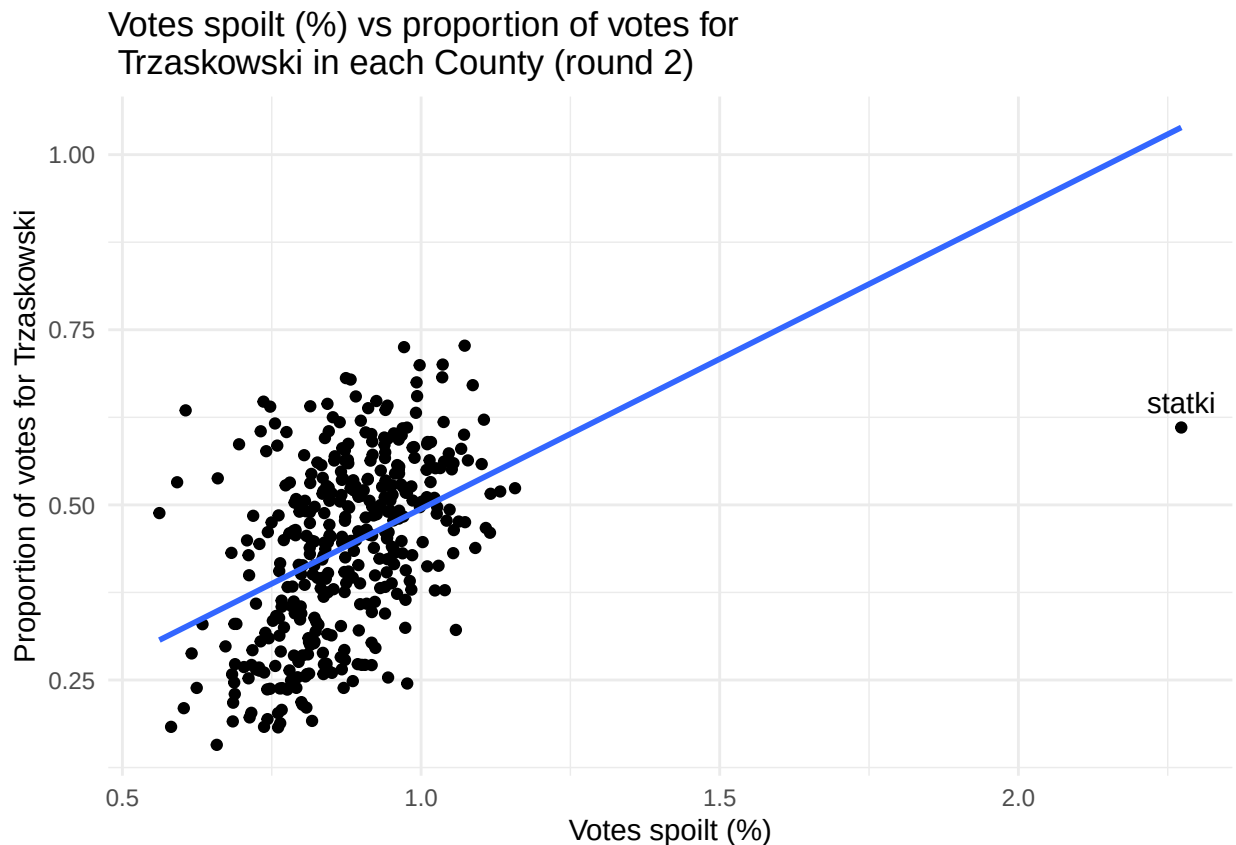
county	pct_spoilt
Skierniewice	1.132525
średzki	1.116488
mogileński	1.115415

```

polpres_spoilt_votes_statki <- polpres_spoilt_votes %>%
  filter(county == "statki")

polpres_spoilt_votes %>%
  ggplot(aes(x = pct_spoilt, y = T2_proportion)) +
  geom_point() +
  geom_text(data = polpres_spoilt_votes_statki,
    aes(label = county),
    vjust = -0.7) + # geom_text used from data science module
  geom_smooth(method = lm, se = FALSE) +
  labs(
    title = "Votes spoilt (%) vs proportion of votes for \n Trzaskowski in each County (round 2)",
    x = "Votes spoilt (%)", y = "Proportion of votes for Trzaskowski") +
  theme_minimal()

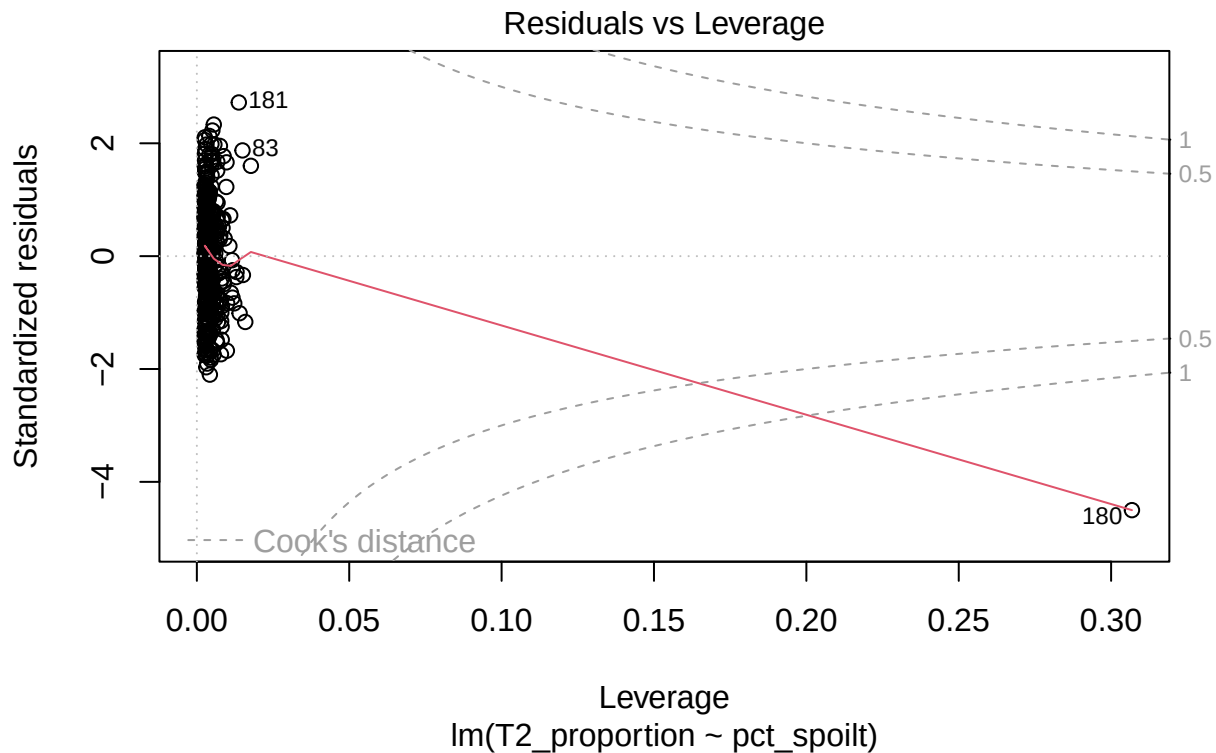
```



I computed the % votes spoilt per county, and then plotted against the proportion of votes for Trzaskowski in r2. From the output, clearly statki is unusual.

I fitted a model and checked diagnostics to see if removing Statki is plausible.

```
lmod_spoilt_T2 <- lm(T2_proportion ~ pct_spoilt, data = polpres_spoilt_votes)
plot(lmod_spoilt_T2, 5)
```



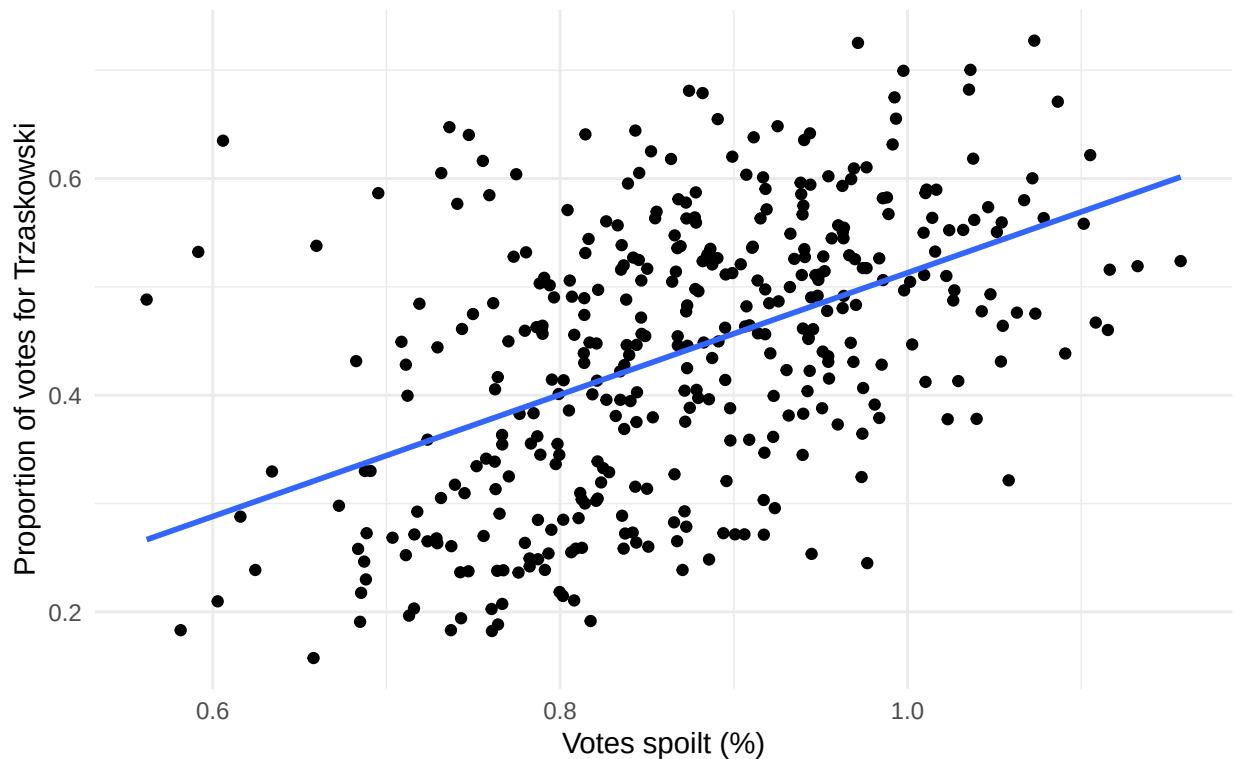
Using residuals vs leverage plot, Statki lies outside Cook's distance so I have decided to remove it before analysis. It would not be sensible to allow Statki with a small relative population to have such leverage over our model estimates.

```
polpres_spoilt_votes2 <- polpres_spoilt_votes %>%
  filter(county != "statki")

lmod_spoilt_T2 <- lm(T2_proportion ~ pct_spoilt, data = polpres_spoilt_votes2)

polpres_spoilt_votes2 %>%
  ggplot(aes(x = pct_spoilt, y = T2_proportion)) +
  geom_point() +
  geom_smooth(method = lm, se = FALSE) +
  labs(
    title = "Votes spoilt (%) vs proportion of votes for \n Trzaskowski in each County (round 2)",
    x = "Votes spoilt (%)", y = "Proportion of votes for Trzaskowski" +
  theme_minimal()
```

Votes spoilt (%) vs proportion of votes for Trzaskowski in each County (round 2)



Having removed “statki” diagnostic plots were fine. The plot suggests some positive linear association between % votes spoilt and proportion for Trzaskowski in r2 in each county.

I investigated more formally:

```
lmod_spoilt_T2_summary <- summary(lmod_spoilt_T2)
```

```
cat("p-value is tiny (",summary(lmod_spoilt_T2)$coefficients["pct_spoilt", 4],"),","\n")
```

```
p-value is tiny ( 1.972991e-23 )
```

```
cat("beta1 estimate is", round(coef(lmod_spoilt_T2)["pct_spoilt"],3),","\n")
```

```
beta1 estimate is 0.562
```

```
cat("R squared value is", round(summary(lmod_spoilt_T2)$r.squared,3),","\n")
```

```
R squared value is 0.231
```

```
# genAI was used to extract the relevant values to save having to output the full summary
```

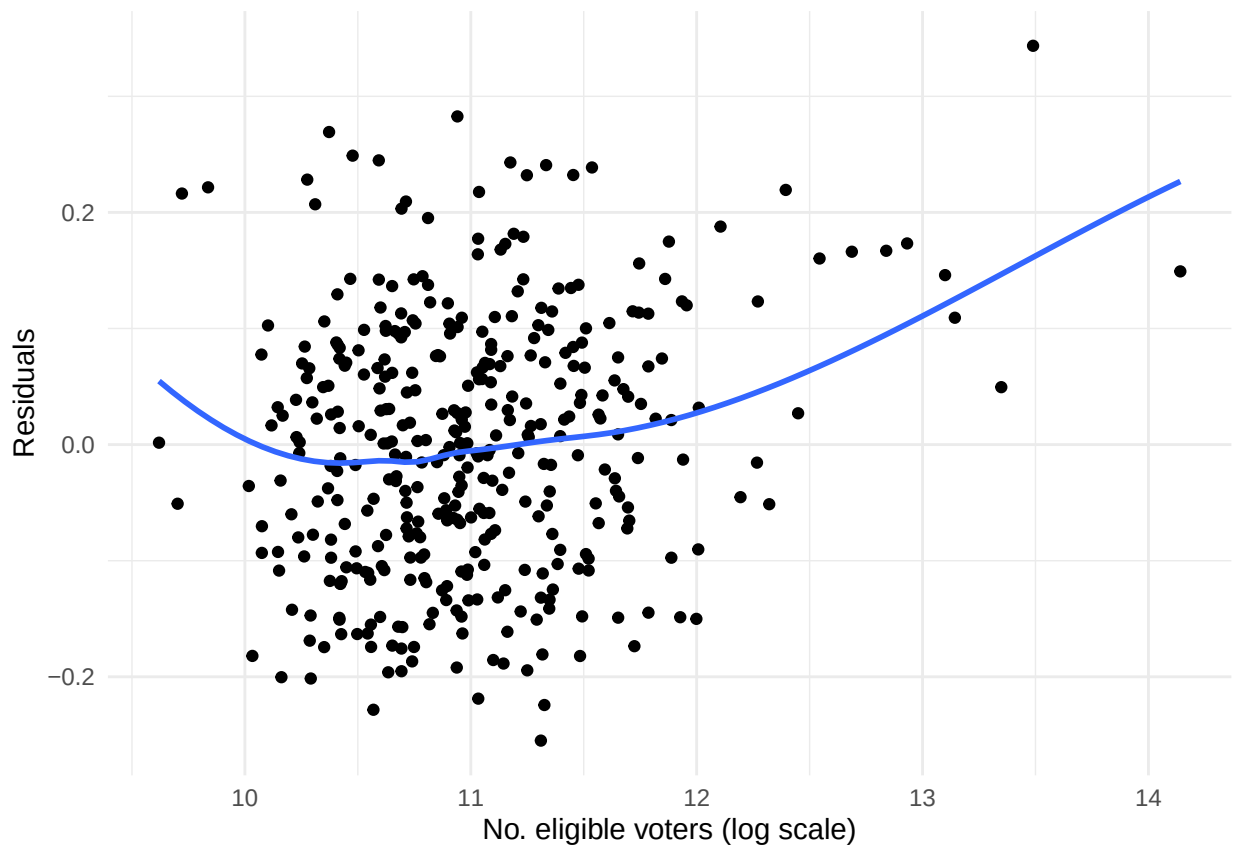
From this summary:

- (1) the p value is extremely small so there is strong statistical evidence of a positive linear association between spoilt vote % and Trzaskowski's vote share.

- (2) β_1 estimate is 0.562, implying a 1 percentage point increase in spoilt results in 0.56218 percentage point increase in proportion of votes for Trzaskowski
- (3) $R^2=0.2311$ which indicates there is still a lot of variation left unexplained between counties i.e. there are other factors which also affect voting proportions in round 2

```
polpres_spoilt_votes2$resid <- resid(lmod_spoilt_T2)

polpres_spoilt_votes2 %>%
  ggplot(aes(y = resid, x = log(eligible2)))+
  geom_point()+
  geom_smooth(se = FALSE)+
  theme_minimal()+
  labs(x = "No. eligible voters (log scale)", y = "Residuals")
```



```
cat("Correlation coefficient between residuals and size is",
    round(cor(log(polpres_spoilt_votes2$eligible2), polpres_spoilt_votes2$resid),3))
```

Correlation coefficient between residuals and size is 0.196

I noted my original model does not account for the size of counties. I decided to take residuals from my model and plot them against no. eligible voters in r2 per county, to investigate whether bigger counties are underestimated by the model. I introduced a log transform on the x axis as the original scale lacked interpretability.

This suggested evidence that Trzaskowski support in larger counties is more likely to be underestimated from %spoilt. Therefore, the previously estimated relationship (between Trzaskowski support and % spoilt) likely understates the overall national effect, because larger counties (with more votes) show stronger than predicted support for Trzaskowski.

I conclude evidence suggests that Trzaskowski had higher support among spoilt votes.

5. USA-style election

```
r2_province_summary <- polpres %>%
  filter(county != "zagranica") %>%
  filter(county != "statki") %>%
  group_by(province) %>%
  summarise(
    eligible_voters = sum(eligible2),
    N2_votes = sum(N2),
    T2_votes = sum(T2))

r2_province_summary %>%
  mutate(winner = ifelse(T2_votes > N2_votes, "Trzaskowski", "Nawrocki")) %>%
  group_by(winner) %>%
  summarise(Electoral_College_Votes = sum(eligible_voters)) %>%
  kable(caption = "No. votes per candidate in a USA-style system")
```

Table 8: No. votes per candidate in a USA-style system

winner	Electoral_College_Votes
Nawrocki	9397732
Trzaskowski	19244178

#genAI used to find ifelse() function to make this code cleaner

```
cat("Trzaskowski wins by",19244178-9397732, "Electoral college votes")
```

Trzaskowski wins by 9846446 Electoral college votes

In the USA electoral system, abroad voters contribute to the state they are registered in. As we have no way of attributing abroad voters to their respective provinces, I have removed “statki” and “zagranica”.

I summarised the round 2 votes for both candidates by province, then created a column indicating who won that province, in order to sum the electoral college votes (ECVs) using “eligible2” variable.

In a USA-style election Trzaskowski wins by around 9.8 million ECVs.

```
N2_deficit <- r2_province_summary %>%
  mutate(N2_deficit = T2_votes - N2_votes) %>%
  filter(N2_deficit>0) %>%
  arrange(N2_deficit)

N2_deficit %>%
  head(8) %>%
  kable(caption = "Provinces ordered by smallest Nawrocki deficit")
```

Table 9: Provinces ordered by smallest Nawrocki deficit

province	eligible_voters	N2_votes	T2_votes	N2_deficit
mazowieckie	4152285	1585019	1602996	17977
warminsko-mazurskie	1035396	320797	343483	22686
opolskie	720842	210321	236992	26671
śląskie	3261536	1111205	1172343	61138
kujawsko-pomorskie	1501949	470660	539860	69200
lubuskie	731055	201982	281477	79495
zachodniopomorskie	1254853	359267	496666	137399
dolnośląskie	2177407	673218	850305	177087

Fewest votes to swing USA-Style election:

I computed provinces with the smallest Nawrocki deficit.

Flipping a province subtracts ECVs from Trzaskowski and adds to Nawrocki, so we only require half the difference: $9846446/2=4923223$. Flipping Mazowieckie and warminsko-mazurskie is a $4152285+1035396=5187681$ ECV swing, so would change the overall outcome. This equates to $(17977+22686)/2=20332$ raw votes change.

These are the 2 provinces with the smallest deficit, and no other provinces have enough ECVs to overturn the result alone, so this is the minimum possible combination.

20332 votes required to change outcome of USA-style election.

```
polpres %>%
  summarise(Nawrocki = sum(N2), Trzaskowski = sum(T2)) %>%
  mutate(Nawrocki_won_by = Nawrocki-Trzaskowski) %>%
  kable(caption= " Votes in the Polish electoral system")
```

Table 10: Votes in the Polish electoral system

Nawrocki	Trzaskowski	Nawrocki_won_by
10606877	10237286	369591

Nawrocki won by 369,591 votes in the Polish system: the minimum votes needed to flip the result is $(369591/2 = 184795.5)$ 184,796.

6. Transfer of first round votes

```
polpres <- polpres %>%
  mutate(
    r1_total = Bartoszewicz + Biejat + Braun + Hołownia + Jakubiak +
      Maciak + Mentzen + Nawrocki + Senyszyn +
      Stanowski + Trzaskowski + Woch + Zandberg,
    Biejat_prop = Biejat / r1_total,
    Zandberg_prop = Zandberg / r1_total,
    Holownia_prop = Hołownia / r1_total,
    Braun_prop = Braun / r1_total,
    Mentzen_prop = Mentzen / r1_total)
```

```

polpres <- polpres %>%
  mutate(T2_proportion = T2/(N2 + T2))

polpres <- polpres %>%
  mutate(
    turnout_r1 = r1_total / eligible,
    turnout_r2 = (N2 + T2 + spoilt) / eligible2,
    late_voters = turnout_r2 - turnout_r1)

lmod_r1 <- lm(
  T2_proportion ~ Biejat_prop + Zandberg_prop + Holownia_prop + Braun_prop + Mentzen_prop + late_voters
  data = polpres)

```

To estimate how voters for Biejat, Zandberg, Hołownia, Braun, and Mentzen and the late joiners voted in round 2 I fitted a linear model with round 2 proportion for Trzaskowski as the response. Initially I fitted it with raw vote counts but the diagnostic plots were quite skewed so I computed the proportion for each candidate instead as the predictors, for which the diagnostics looked good.

```
cat("R squared value is", round(summary(lmod_r1)$r.squared,3), "\n")
```

R squared value is 0.884

```

fstat_lmod_r1 <- summary(lmod_r1)$fstatistic
p_val_lmod_r1 <- pf(fstat_lmod_r1[1], fstat_lmod_r1[2], fstat_lmod_r1[3], lower.tail = FALSE)

cat("Model p-value is significant (", signif(p_val_lmod_r1, 3), ")", "\n")

```

Model p-value is significant (9.66e-172)

```

summary(lmod_r1)$coefficients[,1] %>%
  kable(caption = "Estimated parameters")

```

Table 11: Estimated parameters

	x
(Intercept)	0.5761348
Biejat_prop	5.2793127
Zandberg_prop	-1.0697438
Holownia_prop	2.9058332
Braun_prop	-1.6473285
Mentzen_prop	-1.6514877
late_voters	-1.0547198

From the summary we can deduce:

- (1) r squared is 0.88, which means the model is a very good fit and explains roughly 88% of variation in R2 vote proportion for Trzaskowski. So candidate voted for in round 2 is strongly dependent on round 1 candidate vote.

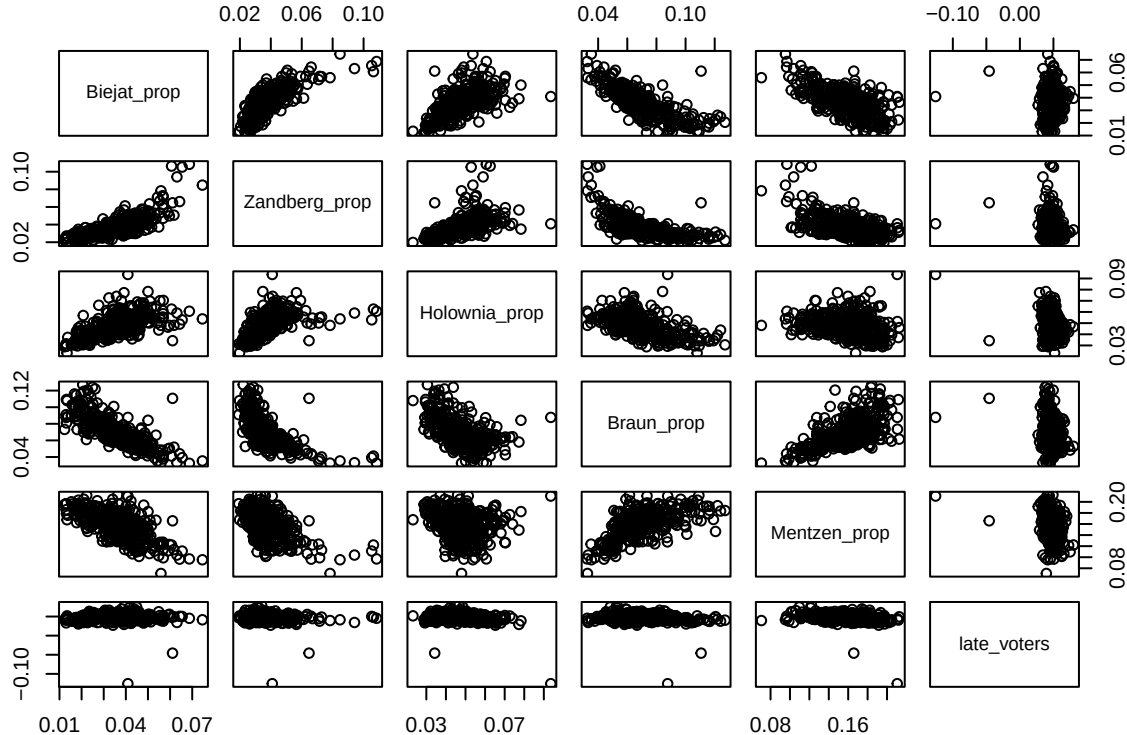
- (2) P value very small so at least one predictor is significant in determining second round outcome
- (3) Biejat coefficient is 5.279, so strongly positive. Indicates counties supporting Biejat in round 1 leikely showed strong support for Trzaskowski in round 2. Same reasoning applies to Holownia.
- (4) The other candidates (and late voters) have negative coefficients and indicate their supporters favoured Nawrocki in R2

In this data, it is very likely that we have collinearity, as candidates can be grouped by political views. I have computed the correlation matrix between predictors below:

```
cor(polpres %>%
  select(Biejat_prop, Zandberg_prop, Holownia_prop, Braun_prop, Mentzen_prop))
```

	Biejat_prop	Zandberg_prop	Holownia_prop	Braun_prop	Mentzen_prop
Biejat_prop	1.0000000	0.7570706	0.6149540	-0.7794629	-0.6512380
Zandberg_prop	0.7570706	1.0000000	0.5409343	-0.6399164	-0.5377407
Holownia_prop	0.6149540	0.5409343	1.0000000	-0.5259494	-0.1973594
Braun_prop	-0.7794629	-0.6399164	-0.5259494	1.0000000	0.6100051
Mentzen_prop	-0.6512380	-0.5377407	-0.1973594	0.6100051	1.0000000

```
pairs(polpres %>%
  select(Biejat_prop, Zandberg_prop, Holownia_prop, Braun_prop, Mentzen_prop, late_voters))
```



```
# genAI was used to find the pairs() function in order to visualise pairwise correlation
```

The correlation matrix suggests counties that had high Biejat proportion also tended to have high Zandberg and Holownia votes, whereas counties with high Mentzen support also had high Braun support. This implies that these groups are ideologically similar. I plotted the pairwise correlations to visualise these relationships. Late voters did not display correlation with any candidates.

Upon further research (sites: Mentzen, Braun, Biejat, Zandberg, 2025 Polish presidential election), I discovered that Mentzen and Braun are right-wing, Biejat and Zandberg are left-wing, and Holownia is centrist. I decided to group the candidates by political views to simplify the model and hopefully increase interpretability.

```
polpres_political_groups <- polpres %>%
  mutate(
    left_p = Biejat_prop + Zandberg_prop,
    centre_p = Holownia_prop,
    right_p = Braun_prop + Mentzen_prop) %>%
  filter(county != "statki") %>%
  filter(county != "zagranica")

lmod_test <- lm(T2_proportion ~ left_p + centre_p + right_p + late_voters, data = polpres_political_groups)
```

I computed the political groups. I found that “statki” and “zagranica” had high leverage in the model (outside cook’s distance), hence I removed them from this model. It is unsurprising they don’t follow the same trends as counties, as the voters are not all in the same geographical location so may have more variation and unpredictability in political views. After removing the diagnostics look good.

```
cat("p-value for late voters is",
    summary(lmod_test)$coefficients["late_voters", "Pr(>|t|)"], "\n")
```

p-value for late voters is 0.9455963

```
drop1(lmod_test)
```

Single term deletions

Model:

```
T2_proportion ~ left_p + centre_p + right_p + late_voters
```

	Df	Sum of Sq	RSS	AIC
<none>			0.85642	-2306.2
left_p	1	0.04622	0.90263	-2288.2
centre_p	1	0.32990	1.18631	-2184.3
right_p	1	0.62811	1.48452	-2099.1
late_voters	1	0.00001	0.85643	-2308.2

The p value for late voters was 0.95, so we could remove it as it is insignificant. The drop1() function confirms that removing late voters lowers the AIC to -2308.2.

```
lmod_r <- lm(T2_proportion ~ left_p + centre_p + right_p, data = polpres_political_groups)

cat("R squared is", summary(lmod_r)$r.squared, "\n")
```

R squared is 0.8590362

```
summary(lmod_r)$coef[,4] %>%  
  kable(caption = "p values")
```

Table 12: p values

	x
(Intercept)	0.0e+00
left_p	5.1e-06
centre_p	0.0e+00
right_p	0.0e+00

#genAI used to extract relevant parts of the summary

```
summary(lmod_r)$coef[,1] %>%  
  kable(caption = "Estimated coefficients")
```

Table 13: Estimated coefficients

	x
(Intercept)	0.6046142
left_p	1.0478840
centre_p	4.2874499
right_p	-1.9680144

Here we deduce:

- (1) $r^2=0.859$ meaning this simplified model still explains most of the variance, despite losing complexity
- (2) All predictors are significant
- (3) Left wing counties have coefficient +1.05 hence counties supporting Biejat and Zandberg showed stronger support for Trzaskowski in R2
- (4) Centrist counties have coefficient +4.23 so counties supporting Holownia showed very strong support for Trzaskowski in R2
- (5) Right wing counties have -1.96 coefficient counties supporting Braun and Mentzen mostly displayed support for Nawrocki in r2
- (6) There is little evidence late voters favoured either candidate in r2